

Lachlan Biersteker

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SUMMARY

Deans list second-year Engineering Physics student at the University of British Columbia with hands-on experience in embedded systems, C/C++ programming, data analysis and machine learning in python, as well as PCB and CAD design. Seeking an engineering internship to apply physics and programming fundamentals to impactful, real-world technical problems.

EDUCATION

University of British Columbia — Vancouver, BC

Sept 2025 – Present

B.A.Sc., Engineering Physics (Second Year) · Expected Graduation 2030

- First-year average 87.8% (Physics & Math average 91.2%).
- Relevant coursework: Differential & Integral Calculus, Linear Algebra, Programming in C/C++, Quantum Mechanics, Thermodynamics & SHM, Electromagnetism, AC/DC Circuits.

Masters Academy — Calgary, AB

2017 – 2025

High School Diploma · Honours with Distinction (90%+ average, Grades 7–12)

- Award for Excellence in Mathematics (Grade 12).

TECHNICAL SKILLS

Languages: Python, C, C++, Java, MATLAB

Embedded & Hardware: STM32, Nordic, Arduino, I2C/UART sensor integration

Design & CAD: Altium, SolidWorks, KiCAD

Tools: VS Code, Git/GitHub, STM32CubeIDE, IntelliJ IDEA

TECHNICAL PROJECTS

Wearable Motion-Sensing System for Machine Learning Exercise Form Analysis · [GitHub](#)

- Built a wearable device that automatically detects reps in an exercise and tells user how they can correct their form for each rep using 2 machine learning models trained on hours of exercise and IMU data.
- **Microcontroller programmed in C++, signal processing and machine learning developed in python.**
- **Fabricated and performed bring up of custom compact PCB with Bluetooth connection designed in Altium**

STM32 Environmental & Flight Motion Sensing System · [GitHub](#)

- Developed embedded firmware in C using STM32CubeIDE and HAL libraries to interface an IMU and barometric pressure sensor over I2C and UART.
- Acquired, processed, and transmitted real-time sensor data over UART; debugged communication and timing issues across the hardware/firmware stack to produce aviation relevant data such as pitch, roll, yaw, altitude.

Avionic sensor PCB (Altium) · [GitHub](#)

- Successfully integrated various bare chip avionic sensors into a 4 layer mixed signal PCB connected to a STM32 microchip through I2C, applying impedance controlling and an antenna bias network for GPS connection.
- Successfully debugged and ensured consistency with manufacturer datasheet and design constraints.

Tesla Coil — Design & Construction

- Designed and built a Tesla coil using inductive coupling, applying electromagnetics, transistor, and high-voltage circuit principles. Carried out full assembly, testing, and troubleshooting.

Model Rocketry — Canadian Association of Rocketry

- Built and launched model rockets; used electronic altimeters and recovery systems to monitor flight performance.

EXPERIENCE & LEADERSHIP

Ski Instructor — WinSport, Calgary, AB

Winter 2022/23, 2023/24

- Taught skiing to ages 3–18+, enforcing safety protocols and adapting to dynamic conditions; CSIA certified.

Bike Instructor — Pedalheads, Calgary, AB

Summer 2023, 2024

- Instructed children ages 5–12 in cycling and roadway safety, prioritizing adaptability and student safety.

Basketball Coach (Grade 8 Team) — Masters Academy

Dec 2024 – Mar 2025

- Ran practices and coached games, developing players' skills and team strategy; former varsity player.

Golf Course Backshop — Radium Springs Course, Radium, BC

- Provided customer service for guests while managing maintaining 100+ golf carts.

May 2026 - Current

CERTIFICATIONS & INTERESTS

Certifications: Altium PCB design · MATLAB Onramp – MathWorks · CSIA Ski Instructor License · CPR-C & First Aid · HCS 3000.

Interests: Programming/embedded systems, rocketry, weightlifting, golf, basketball, skiing, biking